

JUUKA NIEMELA, Finland

WMO: 027910

Lat: 63.233N

Lon: 29.234E

Elev: 377

StdP: 14.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-19.3	-31.2	1.0	-25.2	-24.5	1.4	-19.3	14.4	25.0	13.4	25.3	0.9	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	80.2	65.5	76.5	63.9	73.0	61.8	68.3	76.5	66.0	73.6	63.9	70.8	5.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.3	71.8	63.0	87.3	69.3	60.8	80.6	67.5	32.8	76.0	30.9	73.5	29.3	70.9	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.7	10.3	DB	-30.0	83.9	7.3	4.4	-35.3	87.0	-39.6	89.5	-43.7	92.0	-49.0	95.2	(4)
(5)			WB	-28.7	70.5	8.5	3.8	-34.8	73.2	-39.8	75.5	-44.5	77.6	-50.7	80.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	11.9	15.9	23.0	34.6	46.8	54.1	60.0	56.3	48.0	36.9	29.9	20.3		
	DBStd	18.55	15.23	14.54	10.68	7.40	7.74	6.83	6.17	6.04	5.95	6.75	8.16	13.52		
	HDD50	5683	1180	956	836	462	154	34	5	15	107	409	604	921		
	HDD65	10407	1645	1376	1301	912	566	331	177	277	510	872	1054	1386		
	CDD50	784	0	0	0	1	54	157	315	209	47	1	0	0		
	CDD65	35	0	0	0	0	1	5	23	6	0	0	0	0		
	CDH74	601	0	0	0	0	55	121	327	98	0	0	0	0		
	CDH80	97	0	0	0	0	3	15	62	17	0	0	0	0		
	WSAvg	4.9	4.3	4.8	5.6	5.1	5.5	5.0	4.2	4.1	4.6	5.2	5.2	5.0		
	PrecAvg	29.0	2.2	1.9	1.7	1.4	1.9	2.9	3.5	3.1	2.6	2.6	2.6	2.5		
(15) Precipitation	PrecMax	36.3	4.2	3.7	3.5	2.7	5.6	6.2	6.1	7.1	5.7	5.3	6.0	4.1		
	PrecMin	24.4	0.7	0.2	0.3	0.4	0.2	0.9	1.4	0.7	0.6	0.7	0.5	1.0		
	PrecStd	3.4	0.7	0.9	0.7	0.6	1.1	1.1	1.4	1.5	1.1	1.1	1.1	0.9		
	PrecStd	3.4	0.7	0.9	0.7	0.6	1.1	1.1	1.4	1.5	1.1	1.1	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	40.2	48.0	63.0	79.8	82.4	85.8	82.0	68.8	54.6	48.4	41.2		
		MCWB	36.1	37.0	38.9	48.2	59.6	64.3	70.1	69.3	60.4	51.3	46.7	40.3		
	2%	DB	34.5	36.6	41.8	55.7	74.4	77.8	81.2	76.3	64.3	51.5	44.2	37.2		
		MCWB	33.7	34.8	35.5	45.7	57.2	63.5	67.4	64.3	57.8	49.1	42.9	36.0		
	5%	DB	33.0	34.4	38.4	51.4	68.9	73.3	77.5	72.3	61.3	49.3	41.6	35.0		
		MCWB	32.4	33.5	34.4	42.4	55.4	60.7	66.0	62.3	56.4	47.4	40.8	34.0		
	10%	DB	31.6	33.0	36.0	47.7	63.8	69.1	73.9	68.9	58.8	47.0	39.1	33.5		
		MCWB	31.1	32.3	33.1	40.3	51.6	58.3	63.8	61.1	55.1	45.3	38.4	32.8		
	0.4%	WB	36.5	37.4	39.8	50.3	62.1	68.5	72.0	70.1	61.4	52.8	46.9	40.6		
		MCDB	37.3	39.3	45.5	59.6	74.6	76.5	80.7	80.0	66.2	54.3	48.1	41.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.9	35.1	36.9	46.5	59.0	65.5	69.5	65.9	59.4	50.0	43.2	36.3		
		MCDB	34.3	36.2	40.5	55.1	70.2	74.9	77.9	73.1	62.9	50.9	44.1	36.9		
	5%	WB	32.4	33.5	35.2	43.1	56.4	62.0	67.4	63.9	57.3	47.6	40.9	34.2		
		MCDB	32.8	34.1	37.6	50.1	67.1	70.0	75.2	70.5	60.4	48.7	41.5	34.6		
	10%	WB	30.9	32.4	33.7	40.7	53.0	59.6	65.4	62.1	55.6	45.2	38.6	33.0		
		MCDB	31.5	33.1	35.6	45.7	62.3	67.1	72.4	67.7	58.2	46.8	39.0	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.2	17.8	18.5	22.0	20.3	19.9	18.9	15.6	10.3	7.2	10.0		
		MCDBR	6.6	7.8	15.8	25.7	33.2	28.1	26.6	25.5	21.3	13.7	8.0	6.5		
		MCWBR	6.6	7.1	11.4	15.8	17.7	14.9	14.3	14.9	15.3	11.7	7.8	6.3		
		MCDBR	6.3	7.5	13.4	22.9	29.6	24.0	23.7	21.9	17.9	11.9	7.8	6.0		
	5% WB	MCWBR	6.4	7.0	10.6	14.7	16.7	13.1	13.6	13.3	13.4	11.0	7.6	6.0		
		MCWBR	6.4	7.0	10.6	14.7	16.7	13.1	13.6	13.3	13.4	11.0	7.6	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.225	0.282	0.310	0.348	0.348	0.345	0.371	0.352	0.330	0.309	0.240	0.209		
	taud		1.990	2.108	2.170	2.297	2.452	2.490	2.431	2.470	2.519	2.398	2.079	1.969		
	Ebn at Noon		122	201	246	263	275	279	266	262	246	199	115	61		
	Edh at Noon		10	21	30	33	30	30	31	28	22	16	9	5		
	RadAvg		58	235	705	1163	1504	1670	1582	1203	662	250	62	24		
(45) All-Sky Solar Radiation	RadStd		7	55	86	135	168	101	162	136	76	42	12	3		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	-335	N/A	N/A	N/A

Nomenclature: See separate page